[Paper 12 continued from p. 75.]

This being premised, consider the symbol

$$\left\{\frac{A\rho\sigma\ \ldots (n)}{\rho_k\sigma_k\ \ldots}\right\}, \tag{4}$$

denoting the sum of all the different terms of the form

$$\pm_r \pm_s\ \ldots\ A\rho_{r_1}\sigma_{s_1}\ \ldots\ A\rho_k\sigma_k\ \ldots, \tag{5}$$

the letters

$$r_1,\ r_2 \ldots r_k;\quad s_1,\ s_2 \ldots s_k;\ \&\text{c.}, \tag{6}$$

denoting any permutations whatever, the same or different, of the series of numbers (2) [and the several combinations of ρ, σ ... being understood as denoting suffixes of the A's]. The number of terms represented by the symbol (5) is evidently

$$(1\cdot 2\ \ldots k)^n. \tag{7}$$

In some cases it will be necessary to leave a certain number of the vertical rows ρ, σ ... unpermuted. This will be represented by writing the mark (†) immediately above the rows in question. So that for instance

$$\left\{\frac{\overset{\dagger}{A}\rho\overset{\dagger}{\sigma}\ \ldots (n)}{\rho_k\sigma_k\ \ldots}\right\}, \tag{8}$$

the number of rows with the (†) being x, denotes the sum of the

$$(1\cdot 2\ \ldots k)^{n-x} \tag{9}$$

terms, of the form

$$\pm_r \pm_s\ \ldots\ A\rho_{r_1}\sigma_{s_1}\ \ldots\ A\rho_k\sigma_k\ \ldots. \tag{10}$$

Then it is obvious, that if all the rows have the mark (†) the notation (8) denotes a single product only, and if the mark (†) be placed over all but one of the rows the notation (8) belongs to a determinant. It is obvious also that we may write

$$\left\{\frac{\overset{\dagger}{A}\rho\overset{\dagger}{\sigma}\ \ldots (n)}{\rho_k\sigma_k\ \ldots}\right\} = \Sigma\ \pm_u \pm_v\ \ldots \left\{\frac{A\rho_{u_1}\sigma_{v_1}\ \ldots\ A\rho_{u_k}\sigma_{v_k}}{}\right\}, \tag{11}$$

where Σ refers to the different permutations,

$$u_1,\ u_2, \ldots u_k;\quad v_1,\ v_2, \ldots v_k;\ \&\text{c.}, \tag{12}$$

which can be formed out of the numbers (2). The equation (11) would still be true, if the mark (†) were placed over any number of the columns ρ, σ ...

Suppose in this equation a single column only is left without the mark (†) on the second side of the equation; the first side is then expressed as the sum of a number

$$(1\cdot 2\ \ldots k)^{n-1},\quad \text{or generally}\quad (1\cdot 2\ \ldots k)^{n-x-1}, \tag{13}$$

of determinants, according as we consider the symbol (4) or the more general one (8). And this may be done in n or $(n-x)$ different ways respectively.

It may be remarked, that the symbol (8) is the same in form as if a single column only had the mark (†) over it; the number n being at the same time reduced from n to $(n-x+1)$: for the marked columns of symbols may be replaced by a single marked column of new symbols. Hence, without loss of generality, the theorems which follow may be stated with reference to a single marked column only.

Suppose the letters

$$\rho_1,\ \rho_2,\ldots\rho_k;\quad \sigma_1,\ \sigma_2,\ldots\sigma_k;\ \text{\&c.} \tag{14}$$

denote certain permutations of

$$\alpha_1,\ \alpha_2,\ldots\alpha_k;\quad \beta_1,\ \beta_2,\ldots\beta_k;\ \text{\&c.},, \tag{15}$$

in such a manner that

$$\rho_1=\alpha_{g_1},\ \rho_2=\alpha_{g_2},\ldots\rho_k=\alpha_{g_k};\quad \sigma_1=\beta_{h_1},\ \sigma_2=\beta_{h_2},\ldots\sigma_k=\beta_{h_k},\ \text{\&c..} \tag{16}$$

Then the two following theorems may be proved:

$$\left\{\frac{A\overset{\dagger}{\rho}\ \sigma\ \ldots(n)}{\rho_k\sigma_k\ \ldots}\right\}=\pm_g\pm_h\ \ldots\left\{\frac{A\overset{\dagger}{\alpha}\ \beta\ \ldots(n)}{\alpha_k\beta_k\ \ldots}\right\}, \tag{17}$$

if n be even; but in the contrary case

$$\left\{\frac{A\overset{\dagger}{\rho}\ \sigma\ \ldots(n)}{\rho_k\sigma_k\ \ldots}\right\}=+\pm_g\ \ldots\left\{\frac{A\overset{\dagger}{\alpha}\ \beta\ \ldots(n)}{\alpha_k\beta_k\ \ldots}\right\}. \tag{18}$$

By means of these, and the equation (11), a fundamental property of the symbol (8) may be demonstrated. We have

$$\left\{\frac{A\rho\sigma\ \ldots(n)}{\rho_k\sigma_k\ \ldots}\right\}=\Sigma\ \pm_g\pm_h\ \ldots\left\{\frac{A\overset{\dagger}{\alpha}\ \beta\ \ldots(n)}{\alpha_k\beta_k\ \ldots}\right\}, \tag{19}$$

which, when n is even, reduces itself by (17) to

$$\left\{\frac{A\overset{\dagger}{\alpha}\overset{\dagger}{\beta}\ \ldots(n)}{\alpha_k\beta_k\ \ldots}\right\}\Sigma(\pm_g 1)=1\cdot 2\ \ldots k\left\{\frac{A\overset{\dagger}{\alpha}\ \beta\ \ldots(n)}{\alpha_k\beta_k\ \ldots}\right\}. \tag{20}$$

But when n is odd, from the equation (18),

$$\left\{\frac{A\overset{\dagger}{\alpha}\ \beta\ \ldots(n)}{\alpha_k\beta_k\ \ldots}\right\}\Sigma(\pm_g 1)=0, \tag{21}$$

since the number of negative and positive values of $\pm_g$ are equal.

But when n is odd, from the equation (18), since the number of negative and positive values of $\pm_g$ are equal,

$$\left\{\frac{A\rho\sigma\ \ldots(n)}{\rho_k\sigma_k\ \ldots}\right\} = \left\{\frac{\overset{\dagger}{A}\alpha\ \beta\ \ldots(n)}{\alpha_k\beta_k\ \ldots}\right\}\Sigma(\pm_g 1) = 0. \tag{21}$$

From the equation (20), it follows that when n is even, the value of a symbol of the form

$$\left\{\frac{\overset{\dagger}{A}\alpha\ \beta\ \ldots(n)}{\alpha_k\beta_k\ \ldots}\right\} \tag{22}$$

is the same, over whichever of the columns α, β ... the mark (†) is placed. To denote this indifference, the preceding quantity is better represented by

$$\left\{\frac{\overset{\dagger}{A}\overset{\dagger}{\alpha}\,\beta\ \ldots(n)}{\alpha_k\beta_k\ \ldots}\right\}, \tag{23}$$

this last form being never employed when n is odd, in which case the same property does not hold. Hence also an ordinary determinant is represented by

$$\left\{\frac{\overset{\dagger}{A}\overset{\dagger}{\alpha}\,\beta\ \ldots}{\alpha_k\beta_k\ \ldots}\right\},\quad \left\{\frac{\overset{\dagger}{A}\ 1\,1\ \ldots}{k\,k}\right\}, \tag{24}$$

the latter form being obviously equally general with the former one.

It is obvious from the equations (17), (18), that the expression (22) vanishes, in the case of n even whenever any two of the symbols α are equivalent, or any two of the symbols β, &c.; but if n be odd, this property holds for the symbols β, &c., but not for the marked ones α. In fact, the interchange of the two equal symbols, in each case, changes the sign of the expression (22), but they evidently leave it unaltered, i.e. the quantity in question must be zero.

Consider now the symbol

$$\left\{\frac{\overset{\dagger}{A}\ 1\,1\ \ldots(2p)}{k\,k}\right\}, \tag{25}$$

which, for shortness, may be denoted by

$$\{A\cdot k\cdot 2p\}. \tag{26}$$

I proceed to prove a theorem, which may be expressed as follows:

$$\{A \cdot k \cdot 2p\} \cdot \{B \cdot k \cdot 2q\} = \{AB \cdot k \cdot 2p + 2q - 2\}, \tag{27}$$

where

$$\overline{AB}_{rs\ldots\, xy\ldots} = \Sigma \cdot A_{rs\ldots l}\, B_{xy\ldots l}, \tag{28}$$

the number of the symbols r, s, ... being obviously $2p-1$, and that of x, y, ... being $2q-1$. The summatory sign Σ refers to l, and denotes the sum of the several terms corresponding to values of l from $l=1$ to $l=k$. Also the theorem would be equally true if l had been placed in any position whatever in the series r, s ... l; and again, in any position whatever in the series x, y ... l, instead of at the end of each of these.

With a very slight modification this may be made to suit the case of an odd number instead of one of the two even numbers $2p$, $2q$; (in fact, it is only necessary to place the mark (†) in $\{AB|\ldots\}$ over the column corresponding to the marked column in $\{A|\ldots\}$, $\{A|\ldots\}$ being the symbol for which the number of columns is odd), but it is inapplicable where the two numbers are odd. Consider the second side of (27); this may be expanded in the form

$$\Sigma \pm_s \ldots \pm_x \pm_y \cdot \overline{AB}_{\rho_1\sigma_1\ldots\, x_1y_1\ldots} \cdots \overline{AB}_{\rho_k\sigma_k\ldots\, x_ky_k\ldots}, \tag{29}$$

where Σ refers to the different quantities $s,\ldots,x,y,\ldots$ as in (11).

Substituting from (28), this becomes

$$\Sigma \cdot \Sigma_l \ldots \Sigma_l\, (\pm_s \ldots \pm_x \pm_y\, A_{\rho_1\sigma_1\ldots l_1} \ldots A_{\rho_k\sigma_k\ldots l_k} \cdot B_{x_1y_1\ldots l_1} \ldots B_{x_ky_k\ldots l_k}). \tag{30}$$

Effecting the summation with respect to x, y ... this becomes

$$\Sigma \cdot \Sigma_l \ldots \Sigma_l \pm_s \ldots A_{\rho_1\sigma_1\ldots l_1} \ldots A_{\rho_k\sigma_k\ldots l_k} \cdot \left\{\frac{\overset{\dagger}{B}\ 1 \ldots l_1}{k\, k\, \ldots l_k}\right\}, \tag{31}$$

Σ now referring to $s,\ldots$ only. The quantity under the sign Σ vanishes if any two of the quantities l are equal, and in the contrary case, we have

$$\left\{\frac{\overset{\dagger}{B}\ 1 \ldots l_1}{k\, k\, \ldots l_k}\right\} = \pm_l \{B \cdot k \cdot 2q\}, \tag{32}$$

which reduces the above to

$$\{B \cdot k \cdot 2q\} \cdot \Sigma \pm_s \ldots \pm_l\ A_{\rho_1\sigma_1\ldots l_1} \ldots A_{\rho_k\sigma_k\ldots l_k}, \tag{33}$$

Σ referring to the quantities $s,\ldots$, and also to the quantities l. And this is evidently equivalent to

$$\{A \cdot k \cdot 2p\}\{B \cdot k \cdot 2q\}, \tag{34}$$

the theorem to be proved. It is obvious that when $p=1$, $q=1$, the equation (27), coincides with the theorem (Θ), quoted in the introduction to this paper.

ON THE THEORY OF LINEAR TRANSFORMATIONS.

[From the Cambridge Mathematical Journal, vol. IV. (1845), pp. 193—209.]

THE following investigations were suggested to me by a very elegant paper on the same subject, published in the Journal by Mr Boole. The following remarkable theorem is there arrived at. If a rational homogeneous function U, of the n^{th} order, with the m variables x, y $\ldots$, be transformed by linear substitutions into a function V of the new variables ξ, $\eta\ldots$; if, moreover, θU expresses the function of the coefficients of U, which, equated to zero, is the result of the elimination of the variables from the series of equations $\partial_x U = 0$, $\partial_y U = 0$, &c., and of course θV the analogous function of the coefficients of V: then $\theta V = E^{\alpha} \cdot \theta U$, where E is the determinant formed by the coefficients of the equations which connect x, y $\ldots$ with ξ, $\eta\ldots$, and $\alpha = (m-1)^{n-1}$[1]. In attempting to demonstrate this very beautiful property, it occurred to me that it might be generalised by considering for the function U, not a homogeneous function of the n^{th} order between m variables, but one of the same order, containing n sets of m variables, and the variables of each set entering linearly. The form which Mr Boole's theorem thus assumes is $\theta V = E'^{\alpha'} \cdot E''^{\alpha''} \ldots E^{(n)\alpha^{(n)}} \cdot \theta U$. This it was easy to demonstrate would be true, if θU satisfied a certain system of partial differential equations. I imagined at first that these would determine the function θU, (supposed, in analogy with Mr Boole's function, to represent the result of the elimination of the variables from $\partial_{x_1} U = 0$, $\partial_{y_1} U = 0$, $\ldots \partial_{x_2} U = 0$, &c.): this I afterwards found was not the case; and thus I was led to a class of functions, including as a particular case the function θU, all of them possessed of the same characteristic property. The system of partial differential equations was without difficulty replaced by a more fundamental system of equations, upon which, assumed as definitions, the theory appears to me naturally to depend; and it is this view of it which I intend partially to develope in the present paper.

[1] The value of α was left undetermined, but Mr Boole has since informed me, he was acquainted with it at the time his paper was written; and has given it in a subsequent paper.

I have already employed the notation

$$\left.\begin{array}{cccc} \alpha, & \beta, & \gamma, & \delta,\dots \\ \alpha', & \beta', & \gamma', & \delta',\dots \\ \alpha'', & \beta'', & \gamma'', & \delta'',\dots \\ \vdots & \vdots & \vdots & \vdots \end{array}\right\} \tag{1}$$

$$\left.\begin{array}{cccc} \alpha, & \beta, & \gamma, & \dots \\ \alpha', & \beta', & \gamma', & \dots \\ \alpha'', & \beta'', & \gamma'', & \dots \\ \vdots & \vdots & \vdots & \end{array}\right\}, \tag{2}$$

(where the number of horizontal rows is less than that of vertical ones) to denote the series of determinants, which can be formed out of the above quantities by selecting any system of vertical rows; these different determinants not being connected together by the sign $+$, or in any other manner, but being looked upon as perfectly separate.

The fundamental theorem for the multiplication of determinants gives, applied to these, the formula

$$\left.\begin{array}{cccc} A, & B, & C, & D,\dots \\ A', & B', & C', & D',\dots \\ A'', & B'', & C'', & D'',\dots \\ \vdots & & & \end{array}\right\} = E \left.\begin{array}{cccc} \alpha, & \beta, & \gamma, & \delta,\dots \\ \alpha', & \beta', & \gamma', & \delta',\dots \\ \alpha'', & \beta'', & \gamma'', & \delta'',\dots \\ \vdots & & & \end{array}\right\}, \tag{3}$$

where

$$\begin{array}{rcl} A & = & \lambda\alpha + \lambda'\alpha' + \lambda''\alpha'' + \dots, \\ B & = & \lambda\beta + \lambda'\beta' + \lambda''\beta'' + \dots, \\ A' & = & \mu\alpha + \mu'\alpha' + \mu''\alpha'' + \dots, \\ B' & = & \mu\beta + \mu'\beta' + \mu''\beta'' + \dots, \\ & \&c. & \end{array}, \tag{4}$$

$$E = \left| \begin{array}{ccc} \lambda, & \mu, & \dots \\ \lambda', & \mu', & \dots \\ \vdots & \vdots & \end{array} \right|, \tag{5}$$

and the meaning of the equation is, that the terms on the first side are equal, each to each, to the terms on the second side.

This preliminary theorem being explained, consider a set of arbitrary coefficients, represented by the general formula

$$rst\dots, \tag{6}$$

in which the number of symbolical letters r, s, $\dots$ is n, and where each of these is supposed to assume all integer values, from 1 to m inclusively.

Let

$$ {}_r1a_{t_1\ldots},\ {}_r1'a_{t'_1\ldots},\ \ldots \tag{7} $$

represent the whole series, taken in any order, in which the first symbolical letter is a. Similarly,

$$ {}_{r_1}a_{s_1t_1\ldots},\ {}_{r'_1}a_{s'_1t'_1\ldots},\ \ldots, \tag{8} $$

the whole series of those in which the second symbolical letter is a, and so on.

Imagine a function u of the coefficients, which is simultaneously of the forms

$$ u = H_p \begin{vmatrix} {}_r1a_{t_1\ldots}, & {}_r1'a_{t'_1\ldots}, & \ldots \\ {}_r2a_{t_2\ldots}, & {}_r2'a_{t'_2\ldots}, & \ldots \\ {}_r3a_{t_3\ldots}, & {}_r3'a_{t'_3\ldots}, & \ldots \\ \vdots & \vdots & \end{vmatrix} $$

$$ u = H_p \begin{vmatrix} {}_{r_1}a_{s_1\ldots}, & {}_{r'_1}a_{s'_1\ldots}, & \ldots \\ {}_{r_2}a_{s_2\ldots}, & {}_{r'_2}a_{s'_2\ldots}, & \ldots \\ {}_{r_3}a_{s_3\ldots}, & {}_{r'_3}a_{s'_3\ldots}, & \ldots \\ \vdots & \vdots & \end{vmatrix} \qquad (A), $$

&c.; in which H_p denotes a rational homogeneous function of the order p. The function H is not necessarily supposed to be the same in the above equations, and in point of fact it will not in general be so. The number of equations is of course $= n$.

The function u, whose properties we proceed to investigate, may conveniently be named a "Hyperdeterminant." Any function satisfying any of the equations (A), without satisfying all of them, will be an "Incomplete Hyperdeterminant." But, considering in the first place such as are complete—

Let $rst\ldots$ be a new set of coefficients connected with the former ones by a system of equations of the form

$$ rst\ldots = \lambda\cdot 1st\ldots + \lambda'\cdot 2st\ldots + \ldots + \lambda^{(m)}\cdot m\,st\ldots, \tag{9} $$

(where the r in $\lambda^{(r)}\ldots$ is not an exponent, but an affix).

Suppose $\mathfrak{u}$ is the same function of these new coefficients that u was of the former ones. Then consider the first of the equations (A) and the equation (3), and writing

$$ L = \begin{vmatrix} \lambda, & \lambda', \ldots \\ \lambda_1, & \lambda'_1, \ldots \\ \vdots & \vdots \end{vmatrix}, \tag{10} $$

we have immediately the equation

$$ \mathfrak{u} = L^p\cdot u. \tag{11} $$

Consider the new set of coefficients

$$ rst\ldots = \mu\cdot r1t\ldots + \mu'\cdot r2t\ldots + \ldots + \mu^{(m)}\cdot r\,m\,t\ldots, \tag{12} $$

and $\mathfrak{u}$ the analogous function of these; then, from the second of the equations (A) and the equation (3), and writing

$$ M = \begin{vmatrix} \mu, & \mu', \ldots \\ \mu_1, & \mu'_1, \ldots \\ \vdots & \vdots \end{vmatrix}, \tag{13} $$

$$ \mathfrak{u} = M^p\cdot u = L^pM^p\cdot u. \tag{14} $$

In like manner, considering the new coefficients $rst\ldots$, where

$$rst\ldots = \nu\cdot rs1\ldots + \nu'\cdot rs2\ldots + \ldots + \nu^{(m)}\cdot rs\,m\ \ldots, \tag{15}$$

the new function $\mathfrak{u}$, and the quantity N given by

$$N = \left|\begin{array}{ll} \nu, & \nu',\ldots \\ \nu_1, & \nu_1',\ldots \\ \vdots & \vdots \end{array}\right|, \tag{16}$$

we have, as before,

$$\mathfrak{u} = N^p\cdot u = L^p M^p N^p\cdot u, \tag{17}$$

or

$$\mathfrak{u} = L^p M^p N^p\cdot u; \tag{18}$$

whence, generally, denoting the last result by u',

$$u' = L^p M^p N^p\ \ldots u. \tag{B, 1}$$

Consider now the function

$$\Sigma\Sigma\Sigma\ \ldots(rst\ldots\ x_r\, y_s\, z_t\ \ldots), \tag{19}$$

where the Σ's refer successively to r, s, t, $\ldots$, and denote summations from 1 to m inclusively. If u be looked upon as a derivative from the above function, we may write

$$u = \partial_r\partial_s\partial_t\ \ldots\ \Sigma\Sigma\Sigma\ \ldots(rst\ldots\ x_r\, y_s\, z_t\ \ldots). \tag{20}$$

Assume

$$x_r = M_r^{(1)}y_1 + M_r^{(2)}y_2 + \ldots + M_r^{(m)}y_m. \tag{21}$$

It is easy to obtain

$$\Sigma\Sigma\Sigma\ \ldots(rst\ldots\ x_r\, y_s\, z_t\ \ldots) = \Sigma\Sigma\Sigma\ \ldots(rst\ldots\ X_r\, Y_s\, Z_t\ \ldots), \tag{22}$$

and the formula for (u) becomes

$$\partial_x\partial_y\partial_z\ \ldots\ \Sigma\Sigma\Sigma\ \ldots(rst\ldots\ x_r\, y_s\, z_t\ \ldots) = L^p M^p N^p\ \ldots\partial_X\partial_Y\partial_Z\ \ldots\ \Sigma\Sigma\Sigma\ \ldots(rst\ldots\ X_r\, Y_s\, Z_t\ \ldots). \tag{B, 2}$$

Proceeding to obtain the expression of the coefficients $\mathfrak{r}st\ldots$ in terms of the coefficients $rst\ldots$, we have

$$\mathfrak{r}st\ldots = \Sigma\Sigma\ \ldots(\lambda^f\, \mu^g\, \nu^h\ \ldots\ fgh\ \ldots), \tag{C}$$

where the Σ's refer successively to f, g, h, $\ldots$, denoting summations from 1 to m inclusively. Having this equation, it is perhaps as well to retain

$$u' = L^p M^p N^p\ \ldots u, \tag{B, 1}$$

instead of $(B, 2)$, that form being principally useful in showing the relation of the function u to the theory of the transformation of functions.

It may immediately be seen, that in the equations (B), (C) we may, if we please, omit any number of the marks of variation (∂), omitting at the same time the corresponding signs Σ, and the corresponding factors of the series L, M, N $\ldots$

Also, if u be such as only to satisfy some of the equations (A); then, if in the same formulae we omit the corresponding marks (∂), summatory signs, and terms of the series L, M, N $\ldots$, the resulting equations are still true.

From the formulae (A) we may obtain the partial differential equations

$$\Sigma\Sigma\ \ldots\ \partial_{r_\alpha st\ldots}\partial_{r_\beta st\ldots}\ \ldots u = 0, \quad \text{or} \quad pu,, \tag{D}$$

according as α is not equal, or is equal, to β; and so on: the summatory signs referring in every case to those of the series r, s, t, ..., which are left variable, and extending from 1 to m inclusively.

To demonstrate this, consider the general form of u, as given by the first of the equations (A). This is evidently composed of a series of terms, each of the form

$$c \cdot P \cdot Q \cdot R \ldots (p \text{ factors}),$$

in which

$$P = \begin{vmatrix} {}_1a_{t_1\ldots}, & {}_1'a_{t_1'\ldots}, & \ldots \ (m \text{ terms}) \\ {}_2a_{t_2\ldots}, & {}_2'a_{t_2'\ldots}, & \ldots \\ {}_3a_{t_3\ldots}, & {}_3'a_{t_3'\ldots}, & \ldots \\ \vdots & \vdots & \end{vmatrix},$$

Q, R, &c. being of the same form; and we have

$$\partial_{r_\alpha st\ldots}P = \begin{vmatrix} {}_1a_{t_1\ldots}, & {}_1'a_{t_1'\ldots}, & \ldots \\ {}_2a_{t_2\ldots}, & {}_2'a_{t_2'\ldots}, & \ldots \\ \partial_{r_\alpha st\ldots}, & \ldots & \end{vmatrix} + \&\text{c.} + \&\text{c.},$$

and

$$\Sigma\Sigma\ \ldots\ \partial_{r_\alpha st\ldots}\partial_{r_\beta st\ldots}P = \begin{vmatrix} {}_1a_{t_1'\ldots}, & {}_1'a_{t_1''\ldots}, & \ldots \ (m \text{ terms}) \\ {}_2a_{t_2'\ldots}, & {}_2'a_{t_2''\ldots}, & \ldots \\ \partial_{r_\alpha st\ldots}, & \ldots & \end{vmatrix} = 0;$$

so that all the terms on the second side of the equation vanish. If, however, $\beta = \alpha$, whence, on the second side, we have

$$cQR\ldots P + cPR\ldots Q + \&\text{c.} = p \cdot cPQR\ldots + \&\text{c.} + \&\text{c.} = pu,$$

or the theorem in question is proved.

In the case of an incomplete hyperdeterminant, the corresponding systems of equations are of course to be omitted. In every case it is from these equations that the form of the function u is to be investigated; they entirely replace the system (A).

A very important case of the general theory is, when we suppose the coefficients $rst\ldots$ to have the property $r's't'\ldots = r''s''t''\ldots$, whenever $r's't'\ldots$ and $r''s''t''\ldots$ denote the same combination of letters; and also that the coefficients λ are equal to the coefficients μ, ν ..., each to each. In this case the coefficients $\mathfrak{r}st\ldots$ have likewise the same property, viz. that $\mathfrak{r}''s''t''\ldots = \mathfrak{r}'s't'\ldots$, whenever $r's't'\ldots$ and $r''s''t''\ldots$ denote the same combination of letters.

The equations $(B,1)$, $(B,2)$, become in this case

$$u' = L^{np} \cdot u, \tag{B, 3}$$

instead of $(B,1)$, that form being principally useful in showing the relation of the function u to the theory of the transformation of functions.

It may immediately be seen, that in the equations (B), (C) we may, if we please, omit any number of the marks of variation (∂), omitting at the same time the corresponding signs Σ, and the corresponding factors of the series L, M, N ...

Also, if u be such as only to satisfy some of the equations (A); then, if in the same formulae we omit the corresponding marks (∂), summatory signs, and terms of the series L, M, N ..., the resulting equations are still true.

so that all the terms on the second side of the equation vanish. If, however, $\beta = \alpha$,

$$\Sigma\Sigma \ldots \left(\text{ast} \ldots \frac{d}{d\beta st \ldots}\right) P = P;$$

whence, on the second side of the equation vanish. If, however,

$$cQR \ldots P + cPR \ldots Q + \&\text{c.} = p \cdot cPQR \ldots = pU,$$

or the theorem in question is proved.

In the notation there employed, we have

$$u = \left\{ \begin{array}{c} \overset{\dagger}{1}\overset{\dagger}{1} \ldots (n) \\ \hline 2\,2 \\ \vdots \\ mm \end{array} \right\},$$

a complete hyperdeterminant when n is even; and when n is odd the functions

$$\left\{ \begin{array}{c} \overset{\dagger}{1}\ 1 \ldots (n) \\ \hline 2\,2 \\ \vdots \\ mm \end{array} \right\}, \quad \left\{ \begin{array}{c} 1\ \overset{\dagger}{1} \ldots (n) \\ \hline 2\,2 \\ \vdots \\ mm \end{array} \right\}$$

are each of them incomplete hyperdeterminants.

(A). In the case of $n = 2$, the complete hyperdeterminant is simply the ordinary determinant

$$\begin{vmatrix} 1\,1, & 1\,2, & \ldots & 1\,m \\ 2\,1, & 2\,2, & \ldots & 2\,m \\ \vdots & \vdots & & \vdots \\ m\,1, & m\,2, & \ldots & mm \end{vmatrix}.$$

Stating the general conclusion as applied to this case, which is a very well known one, "If the function

$$U = \Sigma\Sigma\,(rs \cdot x_r y_s),$$

be transformed into a similar function

$$\Sigma\Sigma\,(\mathfrak{rs} \cdot \bar{x}_r \bar{y}_s),$$

by means of the substitutions

$$x_r = \lambda_1^r\,\bar{x}_1 + \lambda_2^r\,\bar{x}_2 + \ldots + \lambda_m^r\,\bar{x}_m,$$

$$y_s = \mu_1^s\,\bar{y}_1 + \mu_2^s\,\bar{y}_2 + \ldots + \mu_m^s\,\bar{y}_s,$$

then

$$\begin{vmatrix} 1\,1, & 1\,2, & \ldots \\ 2\,1, & 2\,2, & \ldots \\ \vdots & & \end{vmatrix} = \begin{vmatrix} \lambda_1^1, & \lambda_2^1, & \ldots \\ \lambda_1^2, & \lambda_2^2, & \ldots \\ \vdots & & \end{vmatrix} \begin{vmatrix} \mu_1^1, & \mu_2^1, & \ldots \\ \mu_1^2, & \mu_2^2, & \ldots \\ \vdots & & \end{vmatrix} \begin{vmatrix} \mathrm{11}, & \mathrm{12}, & \ldots \\ \mathrm{21}, & \mathrm{22}, & \ldots \\ \vdots & & \end{vmatrix}.$$

Also, by what has preceded,

$$\mathfrak{rs} = \Sigma\Sigma\,(\lambda_f^r \cdot \mu_g^s \cdot fg);$$

so that the theorem is easily seen to amount to the following one:—"If the terms of a determinant of the m^{th} order be of the form $\Sigma\Sigma_g(rs, x_r, y_g)$, x, y extending before, from 1 to m inclusively, the determinant itself is the product of three deter- minants; the first formed with the coefficients rs, the second with the quantities x, and the third with the quantities y."

In a following number of the *Journal* I shall prove, and apply to the theories of Maxima and Minima and of Spherical Coordinates, (I may just mention having obtained, in an elegant form, the formulae for transforming from one oblique set of coordinates to another oblique one) the more general theorem.

"If k be the order of the determinant formed as above, the determinant itself is a quadratic function, its coefficients being determinants formed with the coefficients rs, its variables being determinants formed respectively with the variables x and the variables y; and the number of variables in each set being the number of combi- nations of k things out of m, $(= 1$ if $k = m$; if $k > m$ the determinant vanishes)[2]."

I shall give in the same paper the demonstration of a very beautiful theorem, rather relating, however, to determinants than to quadratic functions, proved by Hesse in a Memoir in *Crelle's Journal*, vol. XX. "De curvis et superficiebus secundi ordinis;" and from which he has deduced the most interesting geometrical results. Another Memoir, by the same author, *Crelle*, vol. XXVIII. "Ueber die Elimination der Variabeln aus drei algebraischen Gleichungen vom zweiter Grade mit zwei Variabeln," though relating in point of fact rather to functions of the third order, contains some most important results. A few theorems on quadratic functions, belonging, however, to a different part of the subject, will be found in my paper already quoted in the *Cambridge Philosophical Transactions* [12]; and likewise in a paper in the *Journal*, Chapters in the Algebraical Geometry of n dimensions [11].

I shall, just before concluding this case, write down the particular formula corre- sponding to three variables, and for the symmetrical case. It is, as is well known, the theorem.

"If

$$U = Ax^2 + By^2 + Cz^2 + 2Fyz + 2Gxz + 2Hxy$$

be transformed into

$$\mathfrak{A}\xi^2 + \mathfrak{B}\eta^2 + \mathfrak{C}\zeta^2 + 2\mathfrak{F}\eta\zeta + 2\mathfrak{G}\xi\zeta + 2\mathfrak{H}\xi\eta$$

by means of

$$\begin{aligned} x &= \alpha\xi + \beta\eta + \gamma\zeta, \\ y &= \alpha'\xi + \beta'\eta + \gamma'\zeta, \\ z &= \alpha''\xi + \beta''\eta + \gamma''\zeta, \end{aligned}$$

then

$$(\mathfrak{A}\mathfrak{B}\mathfrak{C} - \mathfrak{A}\mathfrak{F}^2 - \mathfrak{B}\mathfrak{G}^2 - \mathfrak{C}\mathfrak{H}^2 + 2\mathfrak{F}\mathfrak{G}\mathfrak{H}) =$$
$$(\alpha\beta'\gamma'' - \alpha\beta''\gamma' + \alpha'\beta''\gamma - \alpha'\beta\gamma'' + \alpha''\beta\gamma' - \alpha''\beta'\gamma)^2(ABC - AF^2 - BG^2 - CH^2 + 2FGH)."$$

(B) Let $n = 3$, and for greater simplicity $m = 2$; write

$$\begin{aligned} a &= 111, & e &= 112, \\ b &= 211, & f &= 212, \\ c &= 121, & g &= 122, \\ d &= 221, & h &= 222, \end{aligned}$$

so that

$$U = a\,x_1y_1z_1 + b\,x_2y_1z_1 + c\,x_1y_2z_1 + d\,x_2y_2z_1 + e\,x_1y_1z_2 + f\,x_2y_1z_2 + g\,x_1y_2z_2 + h\,x_2y_2z_2.$$

[2] See concluding paragraph of this paper.

There is no complete hyperdeterminant (i.e. for $p = 1$), and the incomplete ones are

$$\begin{aligned} ah - bg - cf + de &= u, \\ ah - de - bg + cf &= u_1, \\ ah - cf - de + bg &= u_2, \end{aligned}$$

suppose. Thus, suppose the transforming equations are

$$x_1 = \lambda_1^1\,\bar{x}_1 + \lambda_2^1\,\bar{x}_2, \quad x_2 = \lambda_1^2\,\bar{x}_1 + \lambda_2^2\,\bar{x}_2,$$

$$y_1 = \mu_1^1\,\bar{y}_1 + \mu_2^1\,\bar{y}_2, \quad y_2 = \mu_1^2\,\bar{y}_1 + \mu_2^2\,\bar{y}_2,$$
$$z_1 = \nu_1^1\,\bar{z}_1 + \nu_2^1\,\bar{z}_2, \quad z_2 = \nu_1^2\,\bar{z}_1 + \nu_2^2\,\bar{z}_2;$$

then

$$\begin{aligned} \bar{u}_1 &= (\lambda_1^1\lambda_2^2 - \lambda_2^1\lambda_1^2)(\mu_1^1\mu_2^2 - \mu_2^1\mu_1^2)(\nu_1^1\nu_2^2 - \nu_2^1\nu_1^2)\,u_1, \quad \text{where } y,\ z \text{ are changed.} \\ \bar{u}_2 &= (\lambda_1^1\lambda_2^2 - \lambda_2^1\lambda_1^2)(\mu_1^1\mu_2^2 - \mu_2^1\mu_1^2)(\nu_1^1\nu_2^2 - \nu_2^1\nu_1^2)\,u_2, \quad z,\ x. \\ \bar{u}\ \ &= (\lambda_1^1\lambda_2^2 - \lambda_2^1\lambda_1^2)(\mu_1^1\mu_2^2 - \mu_2^1\mu_1^2)(\nu_1^1\nu_2^2 - \nu_2^1\nu_1^2)\,u, \quad x,\ y. \end{aligned}$$

We might also have assumed

$$\begin{aligned} u_1 &= ad - bc, \quad eh - gf, \\ u_2 &= af - be, \quad ch - dg, \\ u_3 &= ag - ce, \quad bh - df, \end{aligned}$$

but these are ordinary determinants.

(C) $n = 4$, $m = 2$. Assume

$$\begin{aligned} a &= 1111, & i &= 1112, \\ b &= 2111, & j &= 2112, \\ c &= 1211, & k &= 1212, \\ d &= 2211, & l &= 2212, \\ e &= 1121, & m &= 1122, \\ f &= 2121, & n &= 2122, \\ g &= 2211, & o &= 2212, \\ h &= 2221, & p &= 2222. \end{aligned}$$

$$\begin{aligned} U = {} & a\,x_1y_1z_1w_1 + b\,x_2y_1z_1w_1 + c\,x_1y_2z_1w_1 + d\,x_2y_2z_1w_1 \\ & + e\,x_1y_1z_2w_1 + f\,x_2y_1z_2w_1 + g\,x_1y_2z_2w_1 + h\,x_2y_2z_2w_1 \\ & + i\,x_1y_1z_1w_2 + j\,x_2y_1z_1w_2 + k\,x_1y_2z_1w_2 + l\,x_2y_2z_1w_2 \\ & + m\,x_1y_1z_2w_2 + n\,x_2y_1z_2w_2 + o\,x_1y_2z_2w_2 + p\,x_2y_2z_2w_2, \end{aligned}$$

we have

$$u = ap - bo - cn + dm - el + fk + gj - hi.$$

so that, with the same sets of transforming equations as above, and the additional one,

$$w_1 = \rho_1^1\,\bar{w}_1 + \rho_2^1\,\bar{w}_2,$$

$$w_2 = \rho_1^2\,\bar{w}_1 + \rho_2^2\,\bar{w}_2,$$

we have

$$\bar{u} = (\lambda_1^1\lambda_2^2 - \lambda_2^1\lambda_1^2)(\mu_1^1\mu_2^2 - \mu_2^1\mu_1^2)(\nu_1^1\nu_2^2 - \nu_2^1\nu_1^2)(\rho_1^1\rho_2^2 - \rho_2^1\rho_1^2)\,u;$$

this is important when viewed in reference to a result which will presently be obtained.

If we take the symmetrical case, we have

$$U = ax^4 + 4\beta x^3y + 6\gamma x^2y^2 + 4\delta xy^3 + \epsilon y^4;$$

which is transformed into

$$U = \mathfrak{a}\bar{x}^4 + 4\mathfrak{b}\bar{x}^3\bar{y} + 6\mathfrak{c}\bar{x}^2\bar{y}^2 + 4\mathfrak{d}\bar{x}\bar{y}^3 + \mathfrak{e}\bar{y}^4;$$

by means of

$$x = \lambda\bar{x} + \mu\bar{y},$$

$$y = \lambda'\bar{x} + \mu'\bar{y};$$

then, if

$$u = \alpha\epsilon - 4\beta\delta + 3\gamma^2,$$

$$u' = \mathfrak{ae} - 4\mathfrak{bd} + 3\mathfrak{c}^2,$$

we have

$$u' = (\lambda\mu' - \lambda'\mu)^4 \cdot u.$$

II. Where $p = 2$, $m = 2$, n is odd.

The expression

$$u = \left\{\frac{\overset{\dagger}{1}\ 1\,1\ \ldots(n)}{2\,2\,2\ \ldots}\right\}^2,\quad \left\{\frac{\overset{\dagger}{1}\ 1\,1\,1\ \ldots(n)}{2\,2\,2\,2\ \ldots}\right\}^2,\quad \left\{\frac{1\ \overset{\dagger}{1}\ 1\ \ldots(n)}{2\,2\,2\ \ldots}\right\}^2,\quad \left\{\frac{1\,1\ \overset{\dagger}{1}\ \ldots(n)}{2\,2\,2\ \ldots}\right\}^2$$

is a complete hyperdeterminant; and that over whichever the mark (†) of nonper- mutation is placed. The different expressions so obtained are not, however, all of them independent functions: thus, in the following example, where $n = 3$, the three functions are absolutely identical.

(A). $n = 3$, notation as in I. (B).

$$u = a^2h^2 - b^2g^2 + c^2f^2 + d^2e^2 - 2abhg - 2acfh - 2bdeh - 2cdef + 4adfg + 4bceh,$$

and then

$$\bar{u} = (\lambda_1^1\lambda_2^2 - \lambda_2^1\lambda_1^2)^2(\mu_1^1\mu_2^2 - \mu_2^1\mu_1^2)^2(\nu_1^1\nu_2^2 - \nu_2^1\nu_1^2)^2\,u.$$

This is in many respects an interesting example. We see that the function u may be expressed in the three following forms:

$$u = (ah - bg - cf + de)^2 + 4(ad - bc)(fg - eh), \tag{1}$$

$$u = (ah - bg - de + cf)^2 + 4(af - be)(dg - ch), \tag{2}$$

$$u = (ah - cf - de + bg)^2 + 4(ag - ce)(df - bh). \tag{3}$$

(Pages 91–93 contain a large symbolic expansion of the function for the case $n = 4$, $m = 2$, given as a list of monomials in the coefficients $a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p$ and constants $\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{D}, \mathfrak{E}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H}$. The detailed expansion is reproduced in Cayley's original; here we give the leading expressions in symbolic form.)

We have, as usual,

$$\bar{u} = (\lambda_1^1\lambda_2^2 - \lambda_2^1\lambda_1^2)^2(\mu_1^1\mu_2^2 - \mu_2^1\mu_1^2)^2(\nu_1^1\nu_2^2 - \nu_2^1\nu_1^2)^2(\rho_1^1\rho_2^2 - \rho_2^1\rho_1^2)^2\,u.$$

Particular forms of u are

$$A = 1, \quad B = 0:$$

$$u = (\mathfrak{A} + 3\mathfrak{B} + 3\mathfrak{C} + 6\mathfrak{D}) = (ap - bo - cn + dm - el + fk + gj - hi) = \nu \quad \text{suppose.}$$

$$A = 1, \quad B = 9:$$

$$u = (\mathfrak{A} + 3\mathfrak{B} - 6\mathfrak{C} - 5\mathfrak{D} + 33\mathfrak{E} + 9\mathfrak{F} - 18\mathfrak{G} - 27\mathfrak{H}) = \theta U \quad \text{suppose,}$$

where $\theta U = 0$ is the result of the elimination of the variables from the equations $\partial_x U = 0$, $\partial_y U = 0$, $\partial_z U = 0$, $\partial_w U = 0$, $\partial_{x'} U = 0$, $\partial_{y'} U = 0$, $\partial_{z'} U = 0$, $\partial_{w'} U = 0$. In fact, by an investigation similar to Mr Boole's, applied to a function such as U, it shews that θU has the characteristic property of the function u; also in the present case u is the most general function of this kind, so that θU is obtained from U by properly determining the constant. This has been effected by comparing the value of u, in the symmetrical case, with the value of θU, in the same case, the expanded expression of which is given by Mr Boole in the *Journal*, vol. IV. p. 169. Assuming $A = 1$, the result was $B = 9$. [Incorrect: the result of the elimination is not $\theta U = 0$, but an equation of a higher degree.]

The general form of u now becomes

$$u = ax^2 + \beta\bar{U},$$

in which a, β, are indeterminate.

We have

$$\bar{u} = a\bar{x}^2 + \beta\bar{\bar{U}} = M\,(ax^2 + \beta\bar{U}),$$

where

$$M = (\lambda_1^1\lambda_2^2 - \lambda_2^1\lambda_1^2)^2(\mu_1^1\mu_2^2 - \mu_2^1\mu_1^2)^2(\nu_1^1\nu_2^2 - \nu_2^1\nu_1^2)^2(\rho_1^1\rho_2^2 - \rho_2^1\rho_1^2)^2,$$

and thence

$$\bar{\nu}^2 = M\nu^2,$$

which coincides with a previous formula, and

$$\theta\bar{U} = M\theta U;$$

whence, eliminating M,

$$\frac{\theta\bar{U}}{\bar{\nu}^2} = \frac{\theta U}{\nu^2};$$

an equation which is remarkable as containing only the constants of U and $\bar{U}$; it is an equation of condition which must exist among the constants of $\bar{U}$ in order that this function may be derivable by linear substitutions from U.

In the symmetrical case, or where

$$U = ax^4 + 4\beta x^3 y + 6\gamma x^2 y^2 + 4\delta x y^3 + \epsilon y^4,$$

it has been already seen that ν is given by

$$\nu = \alpha\epsilon - 4\beta\delta + 3\gamma^2.$$

Proceeding to form θU, we have

$$\begin{aligned}
\mathfrak{A} &= \alpha\epsilon^3 - 4\beta\delta^3,\\
\mathfrak{B} &= 4(\alpha\delta^2\epsilon - \alpha\epsilon^2\delta + 3\beta\gamma\epsilon - 3\beta\delta^2\gamma),\\
\mathfrak{C} &= 3(\alpha^2\epsilon^2 + 2\alpha\gamma\epsilon^2 - 4\alpha\delta^2\epsilon),\\
\mathfrak{D} &= 6(\alpha\gamma\epsilon^2 - 2\alpha\delta^2\epsilon + 3\delta^2\gamma\epsilon - 2\delta^2\gamma^2),\\
\mathfrak{E} &= 5\alpha\gamma^3 - 4\beta^3\gamma + \&\text{c.},\\
\mathfrak{F} &= 6(\alpha^2\gamma\epsilon - 2\alpha\beta\delta\epsilon - \alpha\gamma^2\delta + 4\beta^2\gamma\delta - 4\beta\gamma^3 + \&\text{c.}),\\
\mathfrak{G} &= 12(\alpha\beta\delta\gamma - \alpha\beta\delta^2 - \alpha\gamma^2\delta + 2\gamma^3\delta + \beta^2\gamma\delta - 2\beta^3\delta),\\
\mathfrak{H} &= 6(\alpha^2\delta^3 - 3\alpha^2\delta^2\epsilon + 4\alpha\beta\delta\epsilon - 6\beta^2\gamma\delta + \&\text{c.}),
\end{aligned}$$

and these values give

$$\theta U = \alpha^2\epsilon^3 - 6\alpha\beta^2\epsilon^2 - 12\alpha\delta^2\beta\epsilon + 18\alpha\gamma^2\beta^2 + 27\beta^2\delta^4 + 56\delta^2\beta^4\epsilon - 24\beta^4\gamma^3 + \&\text{c.}$$

so that this function, divided by $(\alpha\epsilon - 4\beta\delta + 3\gamma^2)^2$, is invariable for all functions of the fourth order which can be deduced one from the other by linear substitutions. The function $u = 4\beta^2 - 3\gamma^2$ occurs in other investigations; I have met with it in a problem relating to a homogeneous function of two variables, of any order whatever, α, β, γ, δ, ϵ signifying the fourth differential coefficients of the function. But this is only remotely connected with the present subject.

Since writing the above, Mr Boole has pointed out to me that in the transform- ation of a function of the fourth order of the form $\alpha x^4 + 4\beta x^3 y + 6\gamma x^2 y^2 + 4\delta x y^3 + \epsilon y^4$,— besides his function θv, and my quadratic function $\alpha\epsilon - 4\beta\delta + 3\gamma^2$,—there exists a function of the third order $\alpha\epsilon - b^2 - \alpha c^2 - c^3 + 2bcd$, possessing precisely the same characteristic property, and that, moreover, the function θv may be reduced to the form

$$(\alpha\epsilon - 4\beta\delta + 3\gamma^2)^3 - 27(\alpha\epsilon - \alpha\delta^2 - c^3 - c^2 + 2bcd)^2;$$

the latter part of which was verified by trial; the former he has demonstrated in a manner which, though very elegant, does not appear to be the most direct which Mr Boole, in his earliest paper on the subject, *Mathematical Journal*, vol. II. p. 70. The equations $d_x v = 0$, $d_y v = 0$, $d_z v = 0$, imply the corresponding equations for the transformed function: from these equations we might obtain two relations between the coefficients, which, in the case of a function of the fourth order, are of the orders 3 and 4 respectively: these imply the corresponding relations between the coefficients of the transformed function. Let $A = 0$, $B = 0$, $A' = 0$, $B' = 0$, must have $A' = AA + MB$, A, A' being of the order 3 in the coefficients of v, but B, B' only of the third order in the coefficients of v, it is evident that the term MB must disappear, or that the equation is of the form $A' = \lambda A$. The function A is obviously the function which, equated to zero, would imply the elimination of α, β, γ, δ, considered as independent quantities from the equations $\alpha x^4 + 4\beta x^3 y + \&c. = 0$, considered as linear in λ, μ, but λ, μ' being arbitrary, this implies the corresponding equations of the third order, would lead to the result $A = AA$. The function A is obviously the fourth order, for functions of the sixth order, I have actually expanded: but it does not appear to contain the complete theory. Again, in the particular case of homogeneous functions of two variables, the transforming functions may be expressed as symmetrical functions of the roots of the equation $u = 0$, which gives rise to an entirely distinct theory. This, however, I have not as yet developed sufficiently for publication. There does not appear to be anything very directly analogous to the subject of this note, in my general theory: if this be so, it proves the absolute necessity of a distinct investi- gation for the present case, that which I have denominated the symmetrical one.

of the third order $\alpha\epsilon - b\eta - c\theta^2 + 2bcde$, possessing precisely the same characteristic property, and that, moreover, the function θv may be reduced to the form

$$(\alpha\epsilon - 4bd + 3c^2)^3 - 27(\alpha ce - \alpha d^2 - eb^2 - c^3 + 2bcd)^2;$$

the latter part of which was verified by trial; the former he has demonstrated in a manner which, though very elegant, does not appear to be the most direct which can be obtained. The corresponding functions, which in the transformation of ∂v may be considered: from these equations we might obtain two relations between the coefficients, which, in the case of a function of the fourth order, are of the orders 3 and 4 respectively: these imply the corresponding relations between the coefficients of the transformed function. Let $A = 0$, $B = 0$, $A' = 0$, $B' = 0$, imply the corresponding equations that A, λ, μ, but λ, μ' being arbitrary, this implies the corresponding equations of the third order, would lead to the result $A = AA$. The function A is obviously the function which, equated to zero, would express the elimination of α, β, γ, δ, considered as independent quantities from the equations $\alpha x^4 + 4\beta x^3 y + \&\mathrm{c}. = 0$, considered as linear in λ, μ, but λ, μ' being arbitrary, this implies the corresponding equations of the third order, would lead to the result $A = AA$. The function A is obviously the fourth order, for functions of the sixth order, I have actually expanded: but it does not appear to contain the complete theory. Again, in the particular case of homogeneous functions of two variables, the transforming functions may be expressed as symmetrical functions of the roots of the equation $u = 0$, which gives rise to an entirely distinct theory. This, however, I have not as yet developed sufficiently for publication. There does not appear to be anything very directly analogous to the subject of this note, in my general theory: if this be so, it proves the absolute necessity of a distinct investi- gation for the present case, that which I have denominated the symmetrical one.

ON LINEAR TRANSFORMATIONS.

[From the Cambridge and Dublin Mathematical Journal, vol. I. (1846), pp. 104—122.]

In continuing my researches on the present subject, I have been led to a new manner of considering the question, which, at the same time that it is much more general, has the advantage of applying directly to the only case which one can possibly hope to develope with any degree of completeness, that of functions of two variables. In fact the question may be proposed, "To find all the derivatives of any number of functions, which have the property of preserving their form unaltered after any linear transformations of the variables." By Derivative I understand a function deduced in any manner whatever from the given functions, and I give the name of Hyperdeterminant Derivative, or simply of Hyperdeterminant, to those derivatives which have the property just enunciated. These derivatives may easily be expressed explicitly, by means of the known method of the separation of symbols. We thus obtain the most general expression of a hyperdeterminant. But there remains a question to be resolved, which appears to present very great difficulties, that of determining the independent derivatives, and the relation between these and the remaining ones. I have only succeeded in treating a very particular case of this question, which shows however in what way the general problem is to be attacked.

Imagine p series each of m variables

$$x_1,\ y_1,\ldots\&\text{c.}\quad x_2,\ y_2,\ldots\&\text{c.}\quad x_p,\ y_p,\ldots\&\text{c.},$$

where p is at least as great as m.

Similarly p' series each of m' variables

$$x'_1,\ y'_1,\ldots\&\text{c.}\quad x'_2,\ y'_2,\ldots\&\text{c.}\quad \ldots x'_{p'},\ y'_{p'},\ldots\&\text{c.},$$

p' at least as great as m', and so on. Let the analogous variables $\bar{x}$, $\bar{y}$, …be con- nected with these by the equations

$$x = \lambda\bar{x} + \mu\bar{y} + \ldots,$$
$$y = \lambda'\bar{x} + \mu'\bar{y} + \ldots,$$
$$\vdots$$
$$x' = \lambda^1\bar{x}' + \mu^1\bar{y}' + \ldots,$$
$$y' = \lambda'^1\bar{x}' + \mu'^1\bar{y}' + \ldots,$$
$$\vdots$$

where x, y, … or x_1, y_1, … or x_p, y_p, … stand for x_1, y_1, …; x'_p, y'_p, … stand for x'_1, y'_1, …, or $x'_{p'}$, $y'_{p'}$, … or $x'_{p'}$, $y'_{p'}$, &c.… The coefficients λ, μ, …, λ', μ', …, &c.; λ^1, μ^1, …, λ'^1, μ'^1, …remain the same in all these systems. Suppose next,

$$\xi = \delta_x, \quad \eta = \delta_y,$$

i.e.

$$\xi_1 = \delta_{x_1}, \quad \eta_1 = \delta_{y_1}, \ldots, \xi_p = \delta_{x_p}, \ldots$$

(where δ_x, δ_y …are the symbols of differentiation relative to x, y, &c.). Then evidently

$$\xi = \mu'\bar{\xi} + \lambda'\bar{\eta} + \ldots,$$
$$\eta = \mu\bar{\xi} + \lambda\bar{\eta} + \ldots,$$
$$\vdots$$

with similar equations for ξ', η', …Suppose

$$\|\Omega\| = \begin{vmatrix} \xi_1, & \xi_2, & \ldots\xi_p \\ \eta_1, & \eta_2, & \ldots\eta_p \\ \vdots & & \end{vmatrix}, \quad \|\Omega'\| = \begin{vmatrix} \xi'_1, & \xi'_2, & \ldots\xi'_{p'} \\ \eta'_1, & \eta'_2, & \ldots\eta'_{p'} \\ \vdots & & \end{vmatrix}, \ldots$$

that is to say $\|\Omega\|$ is the series of determinants formed by choosing any m vertical columns to compose a determinant, and similarly $\|\Omega'\|$, &c. Suppose, besides,

$$E = \begin{vmatrix} \lambda, & \mu, & \ldots \\ \lambda', & \mu', & \ldots \\ \vdots & & \end{vmatrix}, \quad E' = \begin{vmatrix} \lambda^1, & \mu^1, & \ldots \\ \lambda'^1, & \mu'^1, & \ldots \\ \vdots & & \end{vmatrix}, \ldots$$

Then, by the known properties of determinants,

$$\|\bar{\Omega}\| = E\|\Omega\|, \quad \|\bar{\Omega}'\| = E'\|\Omega'\|, \text{ \&c.},$$

i.e. the terms on the one side are respectively equal to the terms on the other. Hence if

$$\Box = F(\|\Omega\|, \ \|\Omega'\|^{f'}, \ \ldots),$$

i.e. $\Box$ a rational and integral function, homogeneous of the order f in the quantities of the series $\|\Omega\|$, homogeneous of the order f' in the quantities of the series $\|\Omega'\|$, &c., we have immediately

$$\bar{\Box} = E^f E'^{f'} \ldots \Box;$$

or if U be any function whatever of the variables x, $y\ldots$ which is transformed by the linear substitutions above into $\bar{U}$, then

$$\bar{\Box}\bar{U} = E^f E'^{f'} \ldots \Box U,$$

or the function

$$\Box U$$

is by the above definition a hyperdeterminant derivative. The symbol $\Box$ may be called "symbol of hyperdeterminant derivation," or simply "hyperdeterminant symbol."

Let A, B,… represent the different quantities of the series $\|\Omega\|$,—A', B',… those of the series $\|\Omega'\|$, &c.,…, then $\Box$ may be reduced to any single term, and we may write

$$\Box = A^\alpha B^\beta \ldots A'^{\alpha'} B'^{\beta'} \ldots$$

Also U may be supposed of the form

$$U = \Theta\Phi \ldots$$

where Θ, Φ are functions of the variables of one of the sets x, y, …, of one of the sets x', y', &c., thus Θ is of the form

$$F(x,\ y_1, \ldots\ x_2,\ y_2, \ldots\ x_p,\ y_p, \ldots),$$

and so on. The functions Θ, Φ, …may be the same or different. It may be supposed after the differentiations that several of the sets x, y, …, or of the sets x', y', …, become identical; in such cases it will always be assumed that the functions Θ, …into the differentiations enter, are similar; and the differential coefficients absolutely identical, in such cases the variables they contain are made so. Thus the general expression of a hyperdeterminant is placed. The different expressions so obtained are not, however, all of them independent functions: thus, in the following example, where $n = 3$, the three functions are absolutely identical.

What precedes, in the general theory; it might, perhaps have been more clearly by confining it to a particular case: and by doing this from the beginning; it will be seen that it presents no real difficulties. Passing at present to some developments, to do this, I neglect entirely the sets x', y'… and I assume that the number m of variables in each of the sets x, y, … and I assume that the number m of variables in each of the sets x, y, …is equal to two. I consider therefore functions V_1, V_2, V_3 of the variables x_1, y_1, or x_2, y_2, … or x_p, y_p. Writing also

$$\xi_r\eta_s - \xi_s\eta_r = \overline{12},\ \text{\&c.}$$

the symbols A, B, …reduce themselves to $\overline{12}$, $\overline{13}$, …Hence for functions of two variables, there results the following still tolerably general form

$$\Box = \overline{12}^a\overline{13}^b\overline{14}^c \ldots \overline{23}^{a'}\overline{24}^{b'}\overline{34}^{a''} \ldots V_1V_2V_3V_4 \ldots$$

The functions V_1, V_2,... may be the same or different; but they will be supposed the same whenever the corresponding variables are made equal. This equality will be denoted by writing, for instance,

$$\square VV'V''\ldots$$

to represent the value assumed by

$$\square V_1V_2V_3V_4\ldots$$

when after the differentiations

$$x_1, y_1 = x_2, y_2 = x_3, y_3 = x, y,$$

$$x_4, y_4 = x', y';$$

&c.

It is easy to determine the general form of $\square U$. To do this, writing for shortness

$$\alpha + \beta + \gamma + \ldots = f_1,$$

$$\alpha + \beta' + \gamma' + \ldots = f_2,$$

$$\beta + \beta' + \gamma'' + \ldots = f_3,$$

&c.,

$$N = (-)^{a+a'+a''+\ldots+l+l'+l''+\ldots} \frac{(\alpha)!}{(\overline{r})!} \frac{(\beta)!}{(s)!} \frac{(\gamma)!}{(t)!} \cdots \frac{(\delta)!}{(\overline{r}')!} \frac{(\gamma')!}{(s')!} \cdots \frac{(\beta'')!}{(\overline{r}'')!} \cdots$$

$$k^{r+s+t+\ldots} V = \delta_x^r \delta_y^{s+\ldots} V \quad \text{or} \quad V^{r,s},$$

the general term is

$$N \cdot V_1^{f_1} V_2^{f_2} \ldots V_p^{f_p},$$

where r, s, t, ... r', s', r'', ... extend from 0 to s, t, ... β', γ', ... respectively. It would be easy to change this general term in a way similar to that which will be employed presently, for the particular case of $\square V_1V_2V_3$.

If several of the functions become identical, that is when some of the letters f are equivalent, it is clear that the derivative $\square U$ refers to a certain number of functions V_1, V_2,... the same as before, of different, the variables x, y; x', y';... respectively. It that this derivative is homogeneous in the degrees θ_1, θ_1',... with respect to the differential coefficients of the orders f_1, f_1',... (consequently homogeneous in the the order $\theta_1 + \theta_1' + \ldots = p$ suppose. In general, only a single function will be considered, and it will be assumed that $\Box U$ only contains the differential coefficients of the f^{th} order. In this case, the derivative is said to be of the degree p and of the order f. The most convenient classification is by degrees, rather than by orders.

Commencing with the simplest case, that of functions of the second order (and writing V, W instead of V_1, V_2), we have

$$\Box VW = \overline{12}^a\, V'W';$$

(where ξ_1, η_1 apply to V and ξ_2, η_2 to W). This will be constantly represented in the sequel by the notation

$$\overline{12}^a\, VW = B_a\,(V,\ W).$$

Hence, writing

$$\delta_x V = V_r, \quad \delta_{a-r} V = V^{a-r} \ldots,$$

we have

$$B_a\,(V,\ W) = V_a \cdot W_a - \left[\frac{a}{1}\right] V_a^{a-1} \cdot W_{a-1} + \ldots;$$

and in particular, according as a is odd or even,

$$B_a\,(V,\ V) = 0,$$

$$\tfrac{1}{2} B_a\,(V,\ V) = V_a^a - \frac{a}{1} V_a^{a-1} \cdot V_{a-1} + \ldots,$$

continued to the term which contains $V_a^{a/2} V_{a-a/2}$, the coefficient of this last term being divided by two.

Thus, for the functions $\frac{1}{2}(ax^2 + 2bxy + cy^2)$, $\frac{1}{2}(ax^4 + 4bx^3y + 6cx^2y^2 + 4dxy^3 + ey^4)$, &c., if a be made equal to 2, 4, &c. respectively, we have the constant derivatives

$$ac - b^2,$$

$$ae - 4bd + 3c^2,$$

$$ag - 6bf + 15ce - 10d^2,$$

$$ai - 8bh + 28cg - 56df + 35e^2,$$

$$\vdots$$

which have all of them the property of remaining unaltered, *à un facteur près*, when the variables are transformed by means of $x = \lambda\bar{x} + \mu\bar{y}$, $y = \lambda'\bar{x} + \mu'\bar{y}$. Thus, for instance, if these equations give

$$ac - b^2 = (\lambda\mu' - \lambda'\mu)^2\,(ac - b^2);$$

then

$$ac - b^2 = (\lambda\mu' - \lambda'\mu)^2\,(ac - b^2);$$

and so on. This is the general theory, which we call to mind for the case of these constant derivatives.

The above functions may be transformed by means of the identical equation

$$B_n\,(V,\ W) = \overline{12}^{n-2}\,B_2\,(V,\ W),$$

to make use of which, it is only necessary to remark the general formula

$$\xi_1^r \eta_1^s \eta_2^r B_2\,(V,\ W) = B_2\,(\xi^r \eta^s V,\ \xi^r \eta^s W).$$

Thus, if $k = 1$, we obtain for the above series, the new forms

$$\begin{aligned}
&ac - b^2,\\
&\tfrac{1}{2}\{(ae - bd) - 3(bd - c^2)\},\\
&(ag - bf) - 5(bf - ce) + 10(ce - d^2),\\
&\tfrac{1}{2}\{(ai - bh) - 7(bh - cg) + 21(cg - df) - 35(df - e^2)\},
\end{aligned}$$

the law of which is evident. This shows also that these functions may be linearly expressed by means of the series of determinants

$$\begin{vmatrix} a, & b \\ b, & c \end{vmatrix}, \quad \begin{vmatrix} a, & b, & c \\ b, & c, & d \end{vmatrix}, \quad \&\text{c.}$$

We may also immediately deduce from them the derivatives B which relate to two functions. For example, for functions of the sixth order this is

$$ag' + a'g - 6(bf' + b'f) + 15(ce' + c'e) - 20\,dd',$$

which has an obvious connection with

$$ag - 6bf + 15ce - 10d^2;$$

and the same is the case for functions of any order.

The following theorem is easily verified; but I am unacquainted with the general theory to which it belongs.

"If U, V are any functions of the second order, and $W = \lambda U + \mu V$; then

$$B_2'\,[B_2\,(W,\ W),\ B_1\,(W,\ W)] = 0,$$

(where B_2' relates to λ, μ) is the same that would be obtained by the elimination of x, y between $U = 0$, $V = 0$."[3]

In fact this becomes

$$4\,(ac - b^2)(a'c' - b'^2) - (ac' + a'c - 2bb')^2 = 0,$$

which is one of the forms under which the result of the elimination of the variables from two quadratic equations may be written. This is a result for which I am indebted to Mr Boole.

[3]Not given with the present paper.